

Correction scheme Midterm SS 2026

1 Production (1+1+1+1+2+3+5 = 9 Points)

1. a) - d) 1P for each correct constraint
2. e) 1P for mathematical correct constraint. 1P only integer coefficients.
3. f) 1P split max into two constraints. 2P for the abs constraints: either recognize positivity or split the absolute values into two constraints.
4. g)
 - 1P introduce two variables ≥ 0 .
 - 1P require $x_A = \dots$
 - 1P bound one variable by 30.
 - 1P Identify that only the constraint for machine time needs to be updated.
 - 1P update the constraint correctly.

2 Graphical solution of an unbounded LP (6+3+4 +4= 17 Points)

- a)
 - 1P for the trivial constraints $x_i \geq 0$
 - 1P for each correct non-trivial constraint
 - 1P optimal solution
 - 1P optimal value
- b)
 - 1P for the trivial constraints $x_i \geq 0$
 - 1P each for the non-trivial constraints
- c) 1P for each correct pair of objective vector and associated objective value.
- d) Partial points for partial solutions:
 - 1P if one correct vector is given
 - 1P if a correct non-singleton set of vectors is given, e.g., $\{(x_1, x_2) \in \mathbb{R}^2, s.t. x_i \geq 0\}$
 - 1P if the normal vectors n_i are correctly identified.
 - 1P correct set.

3 Simplex Algorithm (3+6+1+1+2+3 = 16 Points)

- a)
 - 1P standard form
 - 1P slack variables
 - 1P canonical form
- b)
 - 1P incoming basic variable
 - 1P outgoing basic variable
 - 1P for each correct row in the simplex tableau. (Alternatively, 1P for each correct column except x_3 x_4)
- c) 1P correct point

- d) 1P correct answer + justification
- e) 1P definition, 1p detection
- f) 1P correct solution, 2P justification via local = global optimality for LPs.

4 Sensitivity Analysis (2 + 3 + 2 + 1 + 3 + 2 = 13 Points)

- a) 2P for correct answer:
 - 1P for "unused resources"
 - 1P for unused butter (s_1) and unused flour (s_2)
- b) 3P for correct answer:
 - 2P for mentioning that the reduced costs must be non-negative
 - 1P for correctly deriving the final conditions $c_1 \geq \frac{1}{2}c_2$ and $c_2 \geq \frac{1}{2}c_1$
- c) 2P for deriving the shadow price for butter: 1P for correctly using $c_2 - \frac{1}{2}c_1$ and 1P for the final shadow price
- d) 1P for correct conclusion (indifference w.r.t. buying additional butter at the given price)
- e) 3P for correct answer:
 - 1P for using s_1 column to derive the range of δ
 - 1P for $2 + \frac{1}{2}\delta \geq 0$ and $4 - \frac{1}{2}\delta \geq 0$
 - 1P for final range of δ
- f) 2P for correct answer:
 - 1P for mentioning that outside this range, one of the basic variables becomes negative
 - 1P for concluding that this implies that the current basis is no longer feasible